Fiscal Year:	FY 2011	**Task Last Updated:**	FY 08/24/2011
PI Name:	Baulch, Janet E. Ph.D.		
Project Title:	Mitigating High Z Radiation Induced Genomic Instability by Non-Protein Thiols		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Irvine	**State:**	CA
Zip Code:	92697	**Congressional District:**	45
Comments:	NOTE: PI previously at University of Maryland; moved to UCA Irvine in summer 2013 (Ed., 11/11/13)		
Project Type:	Ground	**Solicitation / Funding Source:**	2004 Radiation Biology NNH04ZUU005N
Start Date:	07/10/2008	**End Date:**	05/31/2011
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Cucinott1a, Francis	**Contact Phone:**	281-483-0968
Contact Email:	noaccess@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date is 5/31/2011 per S. Krenek/JSC (7/2010). NOTE: Received extension to 5/31/2010 (from 5/31/2009) per J. Dardano/JSC (6/2009)		
Key Personnel Changes/Previous PI:	PI change to Dr. Janet Baulch, effective 7/10/2008. See Morgan, William, for fiscal year reports prior to FY2009.		
COI Name (Institution):			
Grant/Contract No.:	NNJ06HD31G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	That astronauts will be exposed to the space radiation environment is unavoidable, and all radiation exposure has the potential to elicit a carcinogenic response. Cancer is characterized by gross destabilization of the genome. In vivo and in vitro exposure to radiation can lead to genomic instability. This radiation induced genomic instability shares many of the characteristics associated with genetic instabilities observed in human tumors and may well be process that transforms a normal cell to a neoplastic growth following radiation exposure. We have developed a rapid and sensitive assay for measuring induced genomic instability occurring in the progeny of irradiated cells. One way to protect space travelers from the deleterious effects of radiation exposure is with the use of radioprotective compounds. In this application we will investigate the radioprotective effect of three non-protein thiol drugs already approved for clinical use. This will be achieved by investigating three specific aims: Specific aim 1 will test the hypothesis that non-protein thiols can reduce the frequency of genomic instability induced by High Z particles in human cells. These human cells have been transfected with a sensitive and reliable reporter system to analyze delayed recombination/deletion events associated with radiation exposure in the progeny of surviving cells. Specific aim 2 will test the hypothesis that some of the delayed effects of high Z radiation result from nontargeted bystander effects resulting from soluble factors secreted by irradiated cells. Specific aim 3 will investigate the molecular mechanism(s) by which the non-protein thiols exert their protective effect. Specifically we will test the hypothesis that treating cells with these drugs induces elevated intracellular levels of manganese superoxide dismutase and this reduces iron ion induced genomic instability in RKO36 cells. The goal of these studies is to determine whether drugs already approved for use in the human population can reduce genomic instability induced by radiations commonly encountered in space. If so, then these drugs can potentially be taken by astronauts to protect against deleterious effects associated with space radiations.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The basic premise of this application is that astronauts will be exposed to space radiations because there is no way of adequately shielding the spacecraft. Thus, if delayed effects, as manifest as induced genomic instability is a consequence of high LET radiation exposure, and mediated through bystander effects and/or reactive oxygen species, then minimizing the detrimental effects of radiation exposure could be accomplished by the use of effective radical scavengers and antioxidants. While this can be accomplished using classic compounds available in the laboratory situation, e.g., DMSO, BSO etc. there are also a number of drugs used medically in the USA that are effective radical scavengers and/or antioxidants. For example, a number of thiol containing drugs, approved for clinical use e.g., amifostine (WR2721), have well established radioprotective and/or antimutagenic/anticarcinogenic properties. Should we demonstrate that any of these drugs offers statistically significant protection against the detrimental delayed effects of radiation exposure, it is possible that astronauts could take these drugs either prophylactically or immediately after a solar event and decrease the potential health hazards associated with radiation exposure.
Task Progress:	Development of novel radiation protectors During the final year of this project, we conducted preliminary experiments using novel radiation countermeasures. These compounds are of particular interest to us in that they are selectively more cytotoxic to normal MCF10A breast cells than MCF7 breast cancer cells. These compounds selectively sensitize the tumor cells to radiation induced cell killing. Further, it should be noted that two of these novel compounds is approximately as effective as WR-1065 at protecting normal cells from the effects of a 4 Gy dose of radiation, while the cancer cells are at increased susceptibility to radiation induced cell killing. These experiments are almost completed and we hope to submit a manuscript based on this work by the end of 2011.
Bibliography Type:	Description: (Last Updated: 05/01/2017)
Articles in Peer-reviewed Journals	Aypar U, Morgan WF, Baulch JE. "Radiation-induced genomic instability: are epigenetic mechanisms the missing link?" Int J Radiat Biol. 2011 Feb;87(2):179-91. Epub 2010 Nov 2. Review. PMID: 21039330 , Feb-2011
Articles in Peer-reviewed Journals	Aypar U, Morgan WF, Baulch JE. "Radiation-induced epigenetic alterations after low and high LET irradiations." Mutat Res. 2011 Feb 10;707(1-2):24-33. Epub 2010 Dec 13. PMID: 21159317 , Feb-2011
Articles in Peer-reviewed Journals	Goetz W, Morgan MN, Baulch JE. "The effect of radiation quality on genomic DNA methylation profiles in irradiated human cell lines." Radiat Res. 2011 May;175(5):575-87. Epub 2011 Mar 4. PMID: 21375360 , May-2011